Group work problem solving session 3

October 19, 2026

Problem 1: Simplex method

Run Simplex on the following linear program in standard equality form:

$$\max\{x_4 : \ Ax = b, \ x \geq \mathbf{0}\} \tag{P}$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix},$$

with the input feasible basis

$$B = \{1, 2, 3\}.$$

(a) Explain why B is a feasible basis of A. Explain why the given linear program is not in canonical form for this basis.

(b) Using elementary row operations, rewrite the LP into canonical form for the basis B.

(c) How can we tell if this basic feasible solution is optimal? Does the BFS with basis B satisfy the optimality condition?

(d) Complete the remaining steps of Simplex. Use Bland's rule whenever there are multiple choices. (The Bland's rule says that if there are multiple variables that could enter or leave the basis, always choose the one with the smallest index.)

(e) Find a certificate for optimality or unboundedness.

If you would like to practice on another example, move on to Problem 2. Otherwise, move on to Problem 3.

Problem 2: Another example on Simplex

Redo Problem 1 on the following LP

$$\max\{x_4 : \ Ax = b, \ x \geq \mathbf{0}\} \tag{Q}$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 1 & 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix},$$

with input feasible basis

$$B = \{1, 2, 3\}.$$

Problem 3

Let $\alpha \in \mathbb{R}$. Consider the following linear program (P_α) in standard equality form:

$$\max\{c^T x : Ax = b, x \geq \mathbf{0}\},$$

where

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \qquad c = c(\alpha) = \begin{bmatrix} 1 \\ -2 \\ -1 \\ \alpha \end{bmatrix}.$$

(a) Show that (P_α) is feasible for every $\alpha \in \mathbb{R}$.

(b) Express the set of all feasible solutions of (P_α).

(c) Determine all values of α for which (P_α) is unbounded above. For each such α, provide a certificate for unboundedness.

(d) Determine all values of α for which $\bar{x} = (1\ 2\ 1\ 0)^T$ is an optimal solution for (P_α). For each such α, provide a certificate for the optimality of $\bar{x}$.

Solution to Problem 1

(a) The basis matrix is

$$A_B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Since A_B is invertible, B is a basis. Solving $A_B x_B = b$ gives

$$x_B = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

Hence the associated basic solution is

$$\bar{x} = (2, 3, 4, 0, 0)^T,$$

which is feasible. So B is a feasible basis. The LP is not in canonical form for B because $A_B \neq I$.

(b) Apply the row operations

$$R_2 \leftarrow R_2 - R_1, \qquad R_3 \leftarrow R_3 - R_2,$$

where the second operation uses the updated row R_2. This gives

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 & 0 & 4 \end{array}\right].$$

The coefficients for x_B in the objective function are already equal to 0. So we do not need to rewrite the objective function. So the LP in canonical form for B is given by

$$\begin{aligned} \max \quad & (0 \quad 0 \quad 0 \quad 1 \quad 0)^T x \\ \text{s.t.} \quad & \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} x = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \\ & x \geq \mathbf{0}. \end{aligned}$$

(c) A basic feasible solution is optimal if the coefficients $\bar{c}$ in the objective function (written in canonical form) is non-positive.

Since $\bar{c}_4 > 0$, the basic feasible solution with basis B is not optimal.

(d) The variable x_4 enters the basis. The ratio test gives

$$\frac{2}{1} = 2, \qquad \frac{3}{1} = 3, \qquad \frac{4}{1} = 4.$$

Therefore, x_1 leaves the basis. The new basis is

$$B = \{2, 3, 4\}.$$

Rewriting the LP in canonical form for the new basis $B = \{2, 3, 4\}$ we obtain

$$\begin{aligned} \max \quad & (-1 \;\; 0 \;\; 0 \;\; 0 \;\; -1)^T x + 2 \\ \text{s.t.} \quad & \begin{bmatrix} -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \\ & x \geq \mathbf{0}. \end{aligned}$$

The coefficients $\bar{c} = (-1 \; 0 \; 0 \; 0 \; -1)^T \leq \mathbf{0}$. Thus, the new basic feasible solution $(0, 1, 2, 2, 0)^T$ is optimal.

(e) The final basis, written in increasing order, is

$$B = \{2, 3, 4\}.$$

Solving

$$A_B^T y = c_B,$$

we obtain

$$y = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

To see that y certifies the optimality we find We have

$$A^T y = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = c,$$

and

$$b^T y = 2,$$

which is equal to the objective value of the basic feasible solution $(0, 1, 2, 2, 0)^T$.

Solution to Problem 2

(a) Same as part (a) of Problem 1.

(b) Apply the row operations

$$R_2 \leftarrow R_2 - R_1, \qquad R_3 \leftarrow R_3 - R_2,$$

where the second operation uses the updated row R_2. This gives

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 0 & 4 \end{array}\right].$$

The coefficients for x_B in the objective function are already equal to zero. Thus, we do not need to rewrite the objective function.

Therefore, the LP in canonical form for B is

$$\begin{aligned} \max \quad & (0 \;\; 0 \;\; 0 \;\; 1 \;\; 0)^T x \\ \text{s.t.} \quad & \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} x = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \\ & x \geq \mathbf{0}. \end{aligned}$$

(c) A basic feasible solution is optimal if the coefficients $\bar{c}$ in the objective function, written in canonical form, are nonpositive.

Here,

$$\bar{c} = (0 \; 0 \; 0 \; 1 \; 0)^T.$$

Since

$$\bar{c}_4 = 1 > 0,$$

the basic feasible solution with basis B does not satisfy the optimality condition and is not optimal.

(d) By Bland's rule, x_4 enters the basis. The column of x_4 in the canonical constraint matrix is

$$\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}.$$

This column has no positive entry. This implies that the LP is unbounded.

(e) Setting $x_4 = t$ and $x_5 = 0$ we obtain that

$$x_1 = 2 + t, \;\; x_2 = 3, \;\; x_3 = 4.$$

Thus, $x = (2 \; 3 \; 4 \; 0 \; 0)^T + t(1 \; 0 \; 0 \; 1 \; 0)^T$ is a feasible solution for every $t \geq 0$. Let $\bar{x} = (2 \; 3 \; 4 \; 0 \; 0)^T$ and $d = (1 \; 0 \; 0 \; 1 \; 0)^T$. We have that $\bar{x}$ is a feasible solution and

$$Ad = \mathbf{0}, \qquad d \geq \mathbf{0}, \qquad c^T d = 1 > 0.$$

Therefore, $(\bar{x}, d)$ provides a certificate that the LP is unbounded.

Solution to Problem 3

(a) Let

$$\bar{x} = (1, 2, 1, 0)^T.$$

Then $A\bar{x} = b$, and $\bar{x} \geq \mathbf{0}$. Thus, $\bar{x}$ is feasible for every $\alpha \in \mathbb{R}$.

(b) Let $x_4 = t$. The equations imply

$$x_1 = 1 + t, \qquad x_2 = 2 + t, \qquad x_3 = 1 + t.$$

Nonnegativity is equivalent to $t \geq 0$. Therefore, the set of feasible solutions is

$$\left\{ \begin{bmatrix} 1 + t \\ 2 + t \\ 1 + t \\ t \end{bmatrix} : t \geq 0 \right\}.$$

(c) Let

$$x^{(t)} := \begin{bmatrix} 1+t \\ 2+t \\ 1+t \\ t \end{bmatrix}.$$

Then,

$$\begin{aligned} c(\alpha)^T x^{(t)} &= (1+t) - 2(2+t) - (1+t) + \alpha t \\ &= -4 + (\alpha - 2)t. \end{aligned}$$

The LP is unbounded if and only if $\lim_{t\to\infty} c(\alpha)^T x^{(t)} = +\infty$, which is the case exactly when

$$\alpha > 2.$$

For every $\alpha > 2$, a certificate of unboundedness is given by $(\bar{x}, d)$ where

$$\bar{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \qquad d = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

because $\bar{x}$ is a feasible solution and

$$Ad = \mathbf{0}, \qquad d \geq \mathbf{0}, \qquad c^T d = \alpha - 2 > 0.$$

(d) From

$$c(\alpha)^T x^{(t)} = -4 + (\alpha - 2)t,$$

the value is maximized at $t = 0$ if and only if $\alpha < 2$. So for every $\alpha < 2$, $\bar{x}$ is an optimal solution.

Notice that $\bar{x}$ is a BFS for basis $B = \{1, 2, 3\}$. Solving $A_B^T y = c_B$ yields

$$y = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}.$$

Then, y is a certificate for optimality of $\bar{x}$ since

$$A^T y = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix} \geq \begin{bmatrix} 1 \\ -2 \\ -1 \\ \alpha \end{bmatrix} = c$$

as $\alpha \leq 2$. Moreover,

$$b^T y = -4 = c(\alpha)^T \bar{x}.$$